

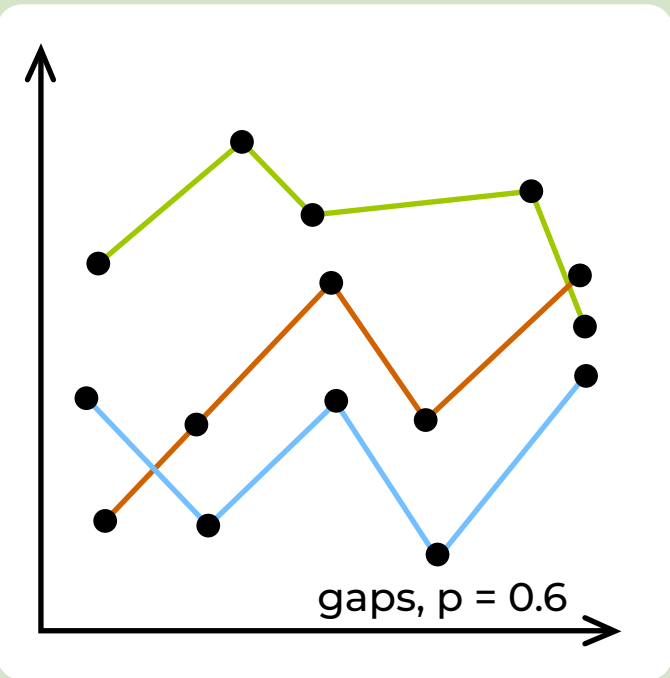
PLACID Pipeline

Zone 1

The Input & Variational Encoding

(The “Anchors”)

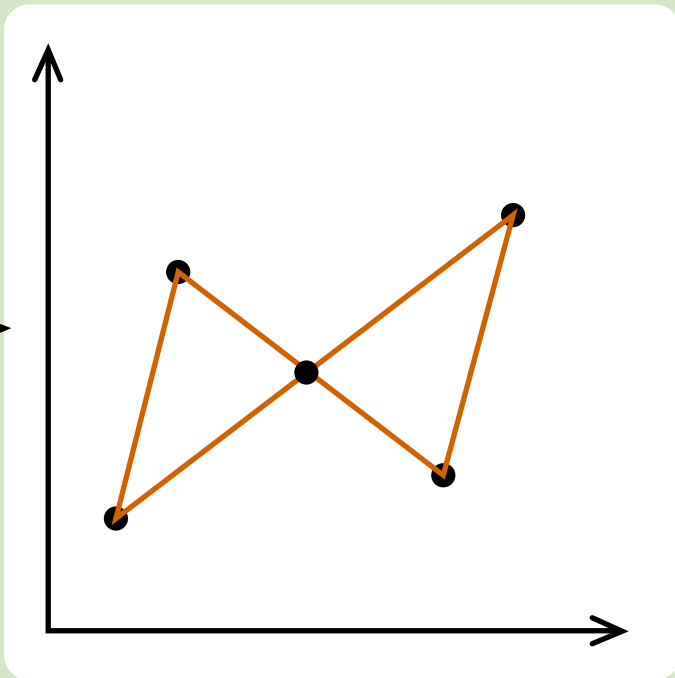
Observation
Space (X)



Encoder

$q\psi(z | x)$

Latent Trajectory
Anchors (Z)



Zone 2

The Latent SDE & Girsanov Control

(The Core Theory)

Latent Causal SDE

Trajectory

$q(z_{tk} | x_{tk})$

$q(z_{tk} | x_{tk})$

Variational Steering Control:
 $u_\phi(t, Z_t, x)$

Invariant
Drift:

$$\text{Total Drift} = \underbrace{\mu_\theta(Z_t)}_{\text{Stationary Structure}} + \underbrace{u_\phi(t, Z_t, x)}_{\text{Transient Forcing}}$$

$$\mathcal{L}_{total} = \underbrace{\text{ELBO}}_{\text{Fit Data}} + \underbrace{\beta \int \|u_\phi\|^2 dt}_{\text{Least Action (Energy Minimization)}}$$

Zone 3

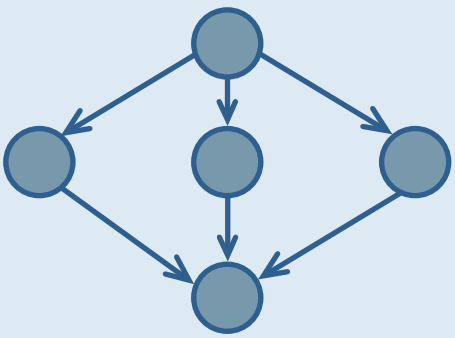
Structural Extraction

(The Output Causal Graph)

Jacobian
Sparsity
($\nabla_z \mu_\theta$)

Proximal
Operator
(Hard Thresholding)

Causal Graph (G)



Final Identified Causal Structure
(DAG/Cyclic Graph)